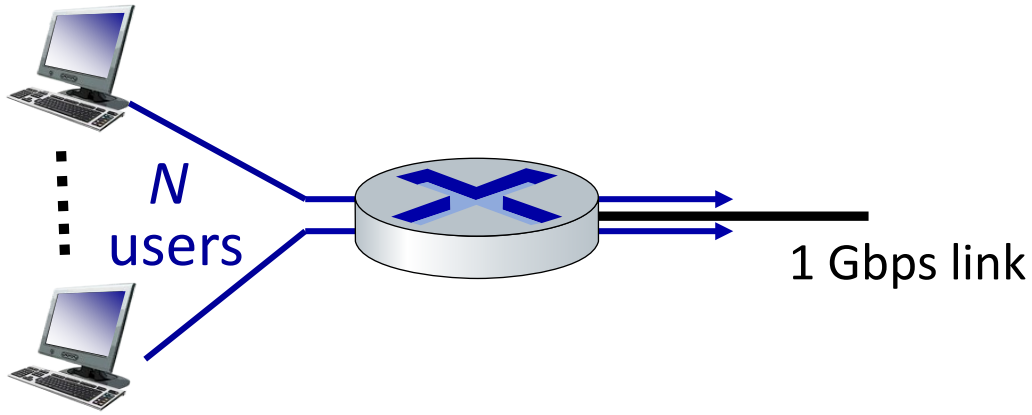
Lecture 4
Packet Switching vs. Circuit Switching
Network Performance Metric

Packet switching versus circuit switching

Packet switching allows more users to use network!

Example:

- 1 Gb/s link
- each user:
  - 100 Mb/s when "active"
  - active 10% of time



- *circuit-switching:* 10 users
- *packet switching:* with 35 users, probability > 10 active at same time is less than .0004

Binomial Probability Distribution

Binomial probability distribution: the likelihood of getting a specific number of successes in a fixed number of independent "yes-or-no trials"

- A fixed number of trials: N
  - e.g., 5 tosses of a coin
- Each try: Binary random variable
  - Head or tail in coin toss
  - Often called as success or failure
  - Prob of success is p, and prob of failure is 1-p
- Consider the example of 5-coin tosses
  - What's the probability that you get exactly 3 heads in 5-coin tosses?

Circuit Switching

Question: Consider a circuit-switched network with N=100 users where each user is independently active with probability p=0.2 and when active, sends data at a rate R=1Mbps. How much capacity must the network provide to guarantee service to all users?

A. 500 Mbps
B. 50 Mbps
C. 100 Mbps
D. 20 Mbps
E. 200 Mbps

Packet Switching

Question: Suppose packet switching is used with N users. What is the probability that a given (specific) user is transmitting, and the remaining users are not transmitting? Each user is active with probability p.

A. $p(1-p)$

B. $p^N(1-p)$

C. $p(1-p^N)$

D. $p(1-p)^{N-1}$

Packet Switching vs. Circuit Switching

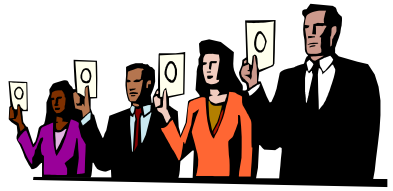
Question: Consider a scenario in which 5 users are being multiplexed over a channel of 10 Mbps. Each user generates traffic at an average rate of 0.21 Mbps, generating traffic at a rate of 15 Mbps when transmitting. Which network architecture works better?

A. Neither works well in this overload scenario
B. Packet switching
C. Circuit switching

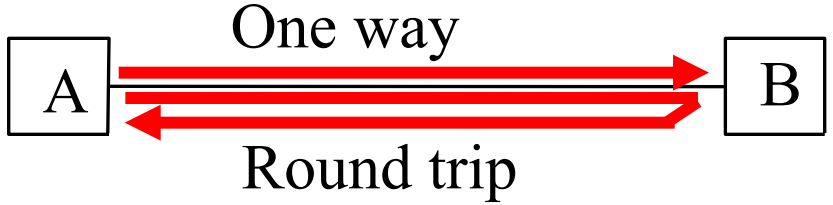
Chapter 1: roadmap

- Basics of Network Building Blocks and Protocols
- Internet Structure
- Network Core: Packet vs. Circuit Switching
- Network Performance: Delay, Packet loss, Throughput
- Protocol Layers and Service Models

Performance Measures



- Metrics to evaluate the performance of the network
- **Goal:** Dimension network (bandwidth, buffers) in order to provide adequate service to users
- **Latency (delay):** Time to send message from point A to point B
  - One-way versus round-trip time (RTT)



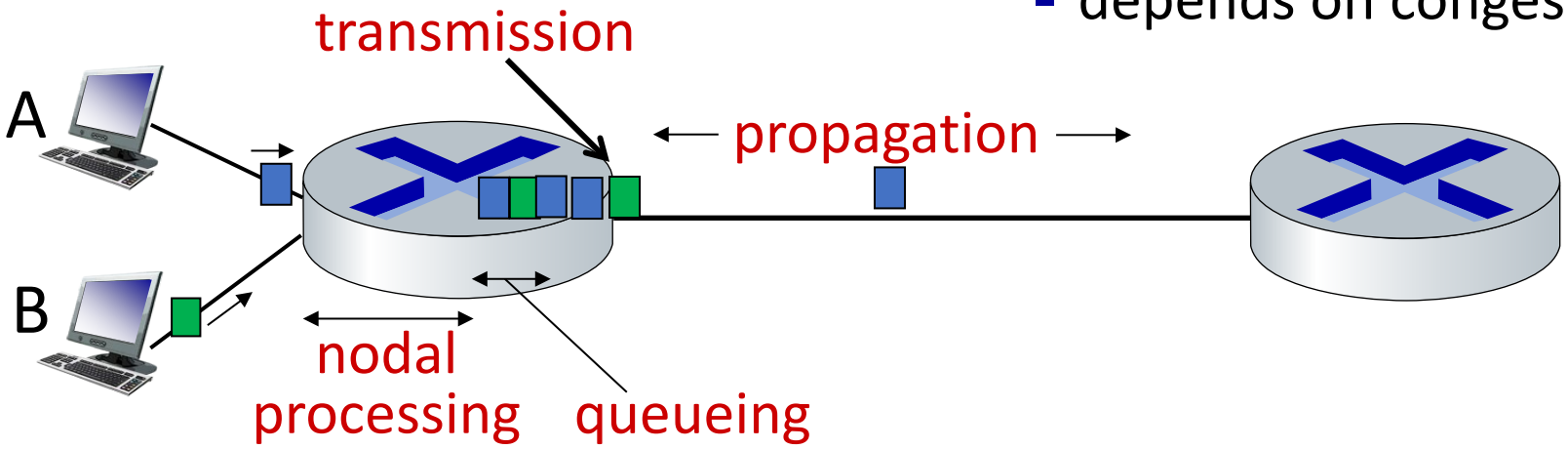
- **Reliability:** number (percentage) of packet loss
- **Throughput (Effective Bandwidth)**
  - The ***effective bandwidth of a link/network*** as measured by a user

Delay components in Packet-Switching Networks

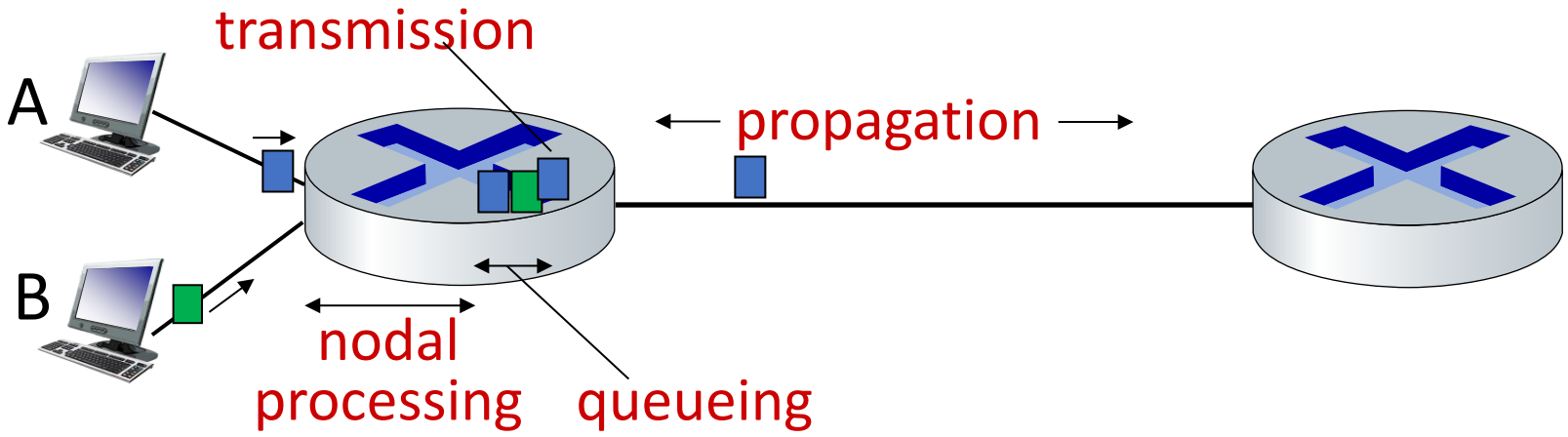
- Packet experience delay on end-to-end path
- Four sources of delay at each hop

d_{proc}: nodal processing
- check bit errors
- determine output link
- typically < msec

d_{queue}: queueing delay
- time waiting at output link for transmission
- depends on congestion level of router



Packet delay: four sources



d_{trans}: transmission delay:

- L: packet length (bits)
- R: link *transmission rate (bps)*
- $d_{trans} = L/R$

d_{prop}: propagation delay:

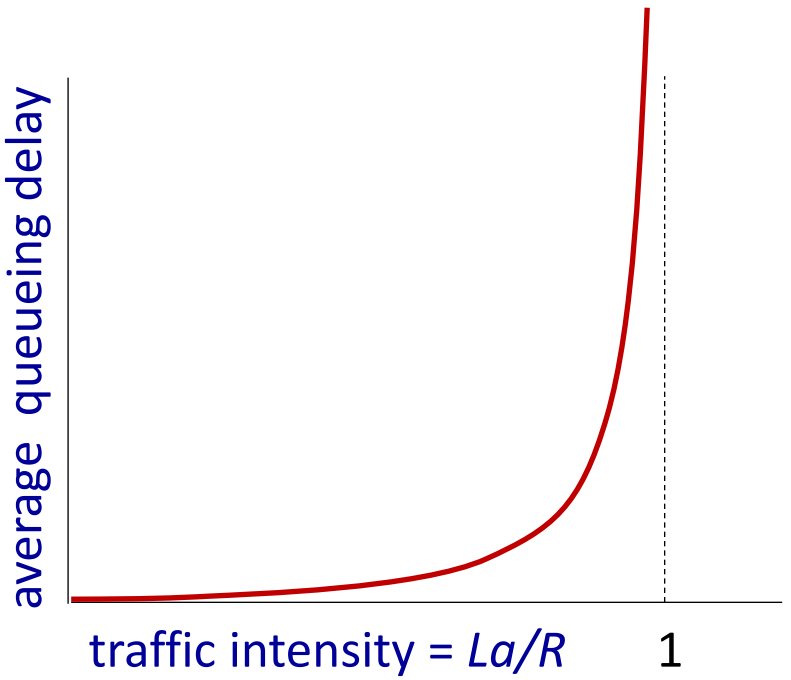
- d: length of physical link
- s: propagation speed (~2×10^8 m/sec)
- $d_{prop} = d/s$

d_{trans} and d_{prop} *very* different

Demo: Link

Packet queueing delay

- *R:* link bandwidth (bps)
- *L:* packet length (bits)
- *a:* average packet arrival rate



- *La/R* ~ 0: avg. queueing delay small
- *La/R* -> 1: avg. queueing delay large
- *La/R* > 1: more "work" is arriving than can be serviced - average delay infinite!



La/R ~ 0

La/R -> 1

Real Internet delays

- What do “real” Internet delay & loss look like?
  - ping
  - traceroute

ping stanford.edu

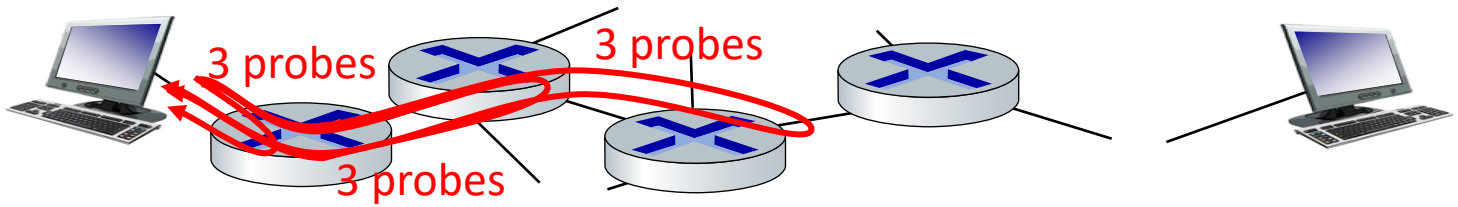
```
C:\Users\mhashemi>ping stanford.edu

Pinging stanford.edu [171.67.215.200] with 32 bytes of data:
Reply from 171.67.215.200: bytes=32 time=42ms TTL=238
Reply from 171.67.215.200: bytes=32 time=42ms TTL=238
Reply from 171.67.215.200: bytes=32 time=42ms TTL=238
Reply from 171.67.215.200: bytes=32 time=43ms TTL=238

Ping statistics for 171.67.215.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 42ms, Maximum = 43ms, Average = 42ms
```

“Real” Internet delays and routes

- traceroute: provides delay measurement from source to router along end-end Internet path towards destination. For all *i:*
  - sends three packets that will reach router *i* on path towards destination (with time-to-live field value of *i*)
  - router *i* will return packets to sender
  - sender measures time interval between transmission and reply



Real Internet delays and routes

Traceroute to stanford.edu

```
C:\Users\mhashemi>tracert stanford.edu

Tracing route to stanford.edu [171.67.215.200]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  gw-vlan110.ittc.ku.edu [129.237.123.254]
  2    <1 ms    <1 ms    <1 ms  bo1.ellx-df-01-1.net.ku.edu [129.237.12.17]
  3     1 ms    <1 ms    <1 ms  vl5.dptvss01.net.ku.edu [129.237.12.3]
  4     1 ms     1 ms    <1 ms  ae5-0.comp-br-01.net.ku.edu [129.237.2.74]
  5     1 ms     1 ms     1 ms  comp-br-01-ae8-untrust.net.ku.edu [129.237.2.14]
  6     2 ms     1 ms     1 ms  kanren-ku-comp-border.peer.net.kanren.net [164.113.216.5]
  7     2 ms     2 ms     2 ms  ptx-bb-kc-walnut-et0-0-0-0.bb.net.kanren.net [164.113.212.14]
  8     5 ms     4 ms     4 ms  fourhundredge-0-0-0-10.2100.core2.kans.net.internet2.edu [64.57.28.177]
  9    41 ms    43 ms    43 ms  fourhundredge-0-0-0-1.4079.core2.denv.net.internet2.edu [163.253.1.250]
 10    41 ms    42 ms    42 ms  fourhundredge-0-0-0-3.4079.core2.salt.net.internet2.edu [163.253.1.169]
 11    43 ms    43 ms    43 ms  fourhundredge-0-0-0-2.4079.core2.sacr.net.internet2.edu [163.253.1.186]
 12    42 ms    40 ms    43 ms  fourhundredge-0-0-0-0.4079.core2.sunn.net.internet2.edu [163.253.1.191]
 13    43 ms    43 ms    47 ms  fourhundredge-0-0-0-22.4079.core1.sunn.net.internet2.edu [163.253.1.24]
 14    40 ms    40 ms    40 ms  hpr-svl-agg10--internet2r&e-100ge.cenic.net [137.164.26.126]
 15    40 ms    41 ms    41 ms  hpr-emvl1-agg-01--svl-agg10--100g.cenic.net [137.164.25.95]
 16    42 ms    42 ms    42 ms  137.164.26.241
 17    41 ms    41 ms    41 ms  campus-ial-nets-b-vl1120.SUNet [171.66.255.232]
 18     *        *        *     Request timed out.
 19    42 ms    42 ms    42 ms  web.stanford.edu [171.67.215.200]

Trace complete.
```

3 delay measurements

KU IP Addresses: 129.237.*.*

* means no response (probe lost, router not replying)

Open Discussion: What is ***Internet2 network?***

Acceptable Network Latencies

- Web Browsing and Cloud Applications
  - Delays under 100 ms is typically acceptable
  - Delays under 50 ms provide very satisfying user experiences.
- Real-time applications: Stock trading, e-commerce, online gaming, etc: < 20-50 ms
- Large data transfers and backups: Delay tolerant

Causes of Network Latency

- Increased Traffic → Congestion → Queueing delay
- Distance → increased number of hops → propagation and processing delay
- Network architecture → inefficient traffic routing → causing delays

Conclusion on Different Dealy Components

Question: Specify whether the following statements are true or false. Select those statements that are ***true*** in iClicker.

A. Transmission delay depends on the size of the packet.
B. Queuing delay depends on the aggregate traffic.
C. Propagation delay is always constant for any two nodes on the Internet.
D. Nodal processing delay does ***NOT*** depend on the router's processor speed.
E. Transmission delay depends on the link capacity.